

VESANTO SONKARI, Finland

WMO: 027100

Lat: 62.922N

Lon: 26.425E

Elev: 120

StdP: 99.89

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-24.1	-30.3	0.2	-27.3	-26.7	0.3	-24.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	27.3	18.6	25.2	17.5	23.3	16.5	20.1	24.4	19.0	23.5	17.9	22.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	21.7	17.5	12.7	20.8	16.3	11.7	19.7	58.2	24.4	54.5	23.5	51.1	22.1	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-29.4	29.6	4.8	2.7	-32.8	31.5	-35.6	33.1	-38.3	34.6	-41.8	36.6		
(5)			-29.6	21.4	4.7	2.1	-33.0	22.9	-35.7	24.1	-38.4	25.3	-41.8	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-10.0	-7.6	-3.7	2.4	9.2	13.2	16.6	14.6	9.8	3.0	-0.5	-5.3		
	DBStd	10.04	7.63	7.37	5.15	3.64	4.63	4.02	3.54	3.22	3.30	3.66	4.30	6.58		
	HDD10.0	2902	619	493	424	227	71	13	1	3	43	217	316	474		
	HDD18.3	5437	877	727	683	477	283	159	78	124	256	474	566	732		
	CDD10.0	546	0	0	0	0	47	111	204	145	38	1	0	0		
	CDD18.3	38	0	0	0	0	1	6	24	7	0	0	0	0		
	CDH23.3	419	0	0	0	0	38	81	238	62	0	0	0	0		
	CDH26.7	76	0	0	0	0	3	12	49	12	0	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	642	40	34	32	34	45	72	91	71	55	61	53	50		
	PrecMax	818	82	65	78	67	115	117	188	136	106	112	115	80		
	PrecMin	528	9	1	4	11	10	20	28	21	23	14	6	15		
	PrecStd	75	19	19	15	16	25	23	35	29	27	27	24	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.4	5.5	9.0	18.3	26.8	28.4	30.1	28.2	20.4	13.3	9.8	5.6		
		MCWB	1.8	3.2	4.0	9.3	15.3	18.3	19.1	19.9	15.3	11.0	8.9	5.2		
	2%	DB	1.2	3.6	6.3	13.8	24.0	25.8	27.9	24.8	18.4	11.2	7.4	3.5		
		MCWB	0.8	2.2	2.7	7.1	13.8	17.3	19.3	17.8	14.5	10.0	6.7	2.9		
	5%	DB	0.4	2.0	4.2	11.2	21.6	23.3	26.1	22.8	16.6	10.1	5.9	2.1		
		MCWB	0.2	1.4	1.6	5.9	13.5	15.9	18.7	16.7	13.4	8.9	5.6	1.6		
	10%	DB	-0.2	0.7	2.7	9.0	18.5	21.2	24.0	20.9	15.1	8.6	4.5	1.2		
		MCWB	-0.4	0.3	0.9	5.2	11.6	15.0	18.1	16.2	12.9	7.6	4.2	0.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	3.6	4.8	9.8	17.2	19.7	22.4	20.9	16.2	11.9	8.9	5.5		
		MCDB	2.5	5.0	7.5	15.1	23.8	25.3	26.2	26.1	19.3	12.9	9.6	5.8		
	2%	WB	0.9	2.4	3.3	8.2	15.3	18.4	20.7	19.0	15.1	10.5	6.8	3.0		
		MCDB	1.1	3.4	5.5	13.2	21.1	23.4	25.2	23.5	17.3	11.2	7.2	3.3		
	5%	WB	0.3	1.4	2.2	6.5	14.0	17.0	19.8	18.0	14.1	9.2	5.6	1.7		
		MCDB	0.4	1.9	3.7	10.3	20.1	22.0	23.9	21.4	16.0	9.9	6.0	2.0		
	10%	WB	-0.4	0.4	1.3	5.3	12.4	15.6	18.8	16.9	13.3	7.7	4.3	0.9		
		MCDB	-0.1	0.8	2.3	8.3	17.7	20.1	23.2	19.6	14.8	8.5	4.5	1.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.8	9.1	9.8	11.2	10.6	10.1	9.7	7.8	5.7	3.5	4.5		
		MCDBR	3.5	5.6	9.6	14.4	16.6	13.8	13.1	12.7	10.6	7.5	5.0	4.2		
		MCWBR	3.5	4.8	6.8	8.5	7.7	6.5	5.8	6.5	7.1	6.0	4.8	4.0		
	5% WB	MCDBR	3.4	5.4	7.9	12.1	14.2	12.9	11.3	10.6	9.2	6.9	5.1	3.9		
		MCWBR	3.3	4.7	6.0	7.7	7.2	6.4	5.8	5.9	6.4	6.0	4.9	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.228	0.284	0.311	0.353	0.348	0.347	0.371	0.355	0.331	0.311	0.246	N/A		
	taud		2.013	2.127	2.199	2.332	2.452	2.481	2.430	2.464	2.519	2.400	2.104	N/A		
	Ebn at Noon		401	643	781	829	869	876	840	823	776	631	376	N/A		
	Edh at Noon		33	66	91	100	97	96	99	88	69	52	29	N/A		
(44) All-Sky Solar Radiation	RadAvg		0.19	0.75	2.27	3.65	4.80	5.25	5.00	3.83	2.15	0.84	0.21	0.08		
	RadStd		0.03	0.17	0.27	0.43	0.52	0.35	0.51	0.43	0.24	0.15	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page